



上海冠显光电科技有限公司  
Shanghai Top Display Optoelectronics Co., LTD

# TR230S DATASHEET

Version 2.2

May 5th, 2024



## Record History

| Version | Date       | Detail                              |
|---------|------------|-------------------------------------|
| V1.0    | 2024-01-04 | First release                       |
| V2.0    | 2024-03-04 | Adding timing and register list     |
| V2.1    | 2024-04-05 | Update register and RGB PIN mapping |
| V2.2    | 2024-05-05 | Update registers                    |
| V2.3    | 2024-11-20 | Update display data writing format  |

## CONTENT

|                                                   |    |
|---------------------------------------------------|----|
| DECLARATION .....                                 | 4  |
| 1.1. GENERAL SPECIFICATION .....                  | 5  |
| 1.2. TR230S BLOCK DIAGRAM .....                   | 7  |
| 1.3. DBU BLOCK .....                              | 8  |
| 1.4. PIN DEFINITION .....                         | 9  |
| 1.4.1. DBU PIN .....                              | 9  |
| 1.4.2. POWER PIN .....                            | 10 |
| 1.4.3. OTHER GPIOs .....                          | 10 |
| 1.4.4. LCD INTERFACE PIN MAPPING .....            | 11 |
| 1.5. INTERFACE DESCRIPTION .....                  | 13 |
| 1.5.1. SPI INTERFACE (SPI-4WIRE, QSPI-4SDA) ..... | 13 |
| 1.5.2. 8080 INTERFACE .....                       | 16 |
| 1.6. REGISTERS .....                              | 19 |
| 1.6.1. Version (01H) .....                        | 19 |
| 1.6.2. Self Test (12H) .....                      | 19 |
| 1.6.3. PWM Duty Ratio Setting (20H) .....         | 19 |
| 1.6.4. PWM Frequency Setting (21H) .....          | 19 |
| 1.6.5. Display Off (28H) .....                    | 19 |
| 1.6.6. Display On (29H) .....                     | 19 |
| 1.6.7. COL_ADR (2AH) .....                        | 19 |
| 1.6.8. ROW_ADR (2BH) .....                        | 20 |
| 1.6.9. DISPaly DATA (2CH) .....                   | 20 |
| 1.6.10. DISPLAY FORMAT (3AH) .....                | 20 |
| 1.6.11. Soft Reset (5AH) .....                    | 20 |
| 1.6.12. RGB Interface Setting (70H) .....         | 20 |
| 1.6.13. Resolution and Clock Setting (71H) .....  | 21 |
| 1.6.14. Horizontal Porch Setting (72H) .....      | 21 |
| 1.6.15. Vertical Porch Setting (73H) .....        | 21 |
| 1.6.16. FONT Property Setting (75H) .....         | 21 |
| 1.6.17. Host SPI function Enable (76H) .....      | 22 |
| 1.6.18. Host SPI Sending format (80H) .....       | 22 |
| 1.6.19. 16*16 Standard FONT library (81H) .....   | 22 |
| 1.6.20. 24*24 Standard FONT library (82H) .....   | 22 |
| 1.6.21. TP FUNCTION ENABLE (A0H) .....            | 23 |
| 1.6.22. TP STATUS (A1H) .....                     | 23 |



---

|                                         |    |
|-----------------------------------------|----|
| 1.6.23. TP Coordinate (A2H) .....       | 23 |
| 1.6.24. Clearing TP STATUS (A3H) .....  | 23 |
| 1.6.25. Drawing LINE (B0H) .....        | 23 |
| 1.6.26. Drawing CIRCLE (B1H) .....      | 24 |
| 1.6.27. Drawing RECTANGLE (B2H) .....   | 24 |
| 1.6.28. MIRROR AND ROTATION (ACH) ..... | 24 |
| 1.7. ELECTRICAL CHARACTERISTICS .....   | 25 |
| 1.8. PACKAGE .....                      | 26 |

---

## DECLARATION

THIS DOCUMENTATION IS THE ORIGINAL WORK AND COPYRIGHTED PROPERTY OF Shanghai Top Display Optoelectronics ( “TDO” ). REPRODUCTION IN WHOLE OR INPART MUST OBTAIN THE WRITTEN APPROVAL OF TDO AND GIVE CLEAR ACKNOWLEDGEMENT TO THE COPYRIGHT OWNER.

THE INFORMATION FURNISHED BY TDO IS BELIEVED TO BE ACCURATE AND RELIABLE. TDO RESERVES THE RIGHT TO MAKE CHANGES IN CIRCUIT DESIGN AND/OR SPECIFICATIONS AT ANY TIME WITHOUT NOTICE. TDO DOES NOT ASSUME ANY RESPONSIBILITY AND LIABILITY FOR ITS USE. NOR FOR ANY INFRINGEMENTS OF PATENTS OR OTHER RIGHTS OF THE THIRD PARTIES WHICH MAY RESULT FROM ITS USE. NO LICENSE IS GRANTED BY IMPLICATION OR OTHERWISE UNDER ANY PATENT OR PATENT RIGHTS OF TDO. THIS DATASHEET NEITHER STATES NOR IMPLIES WARRANTY OF ANY KIND, INCLUDING FITNESS FOR ANY PARTICULAR APPLICATION.

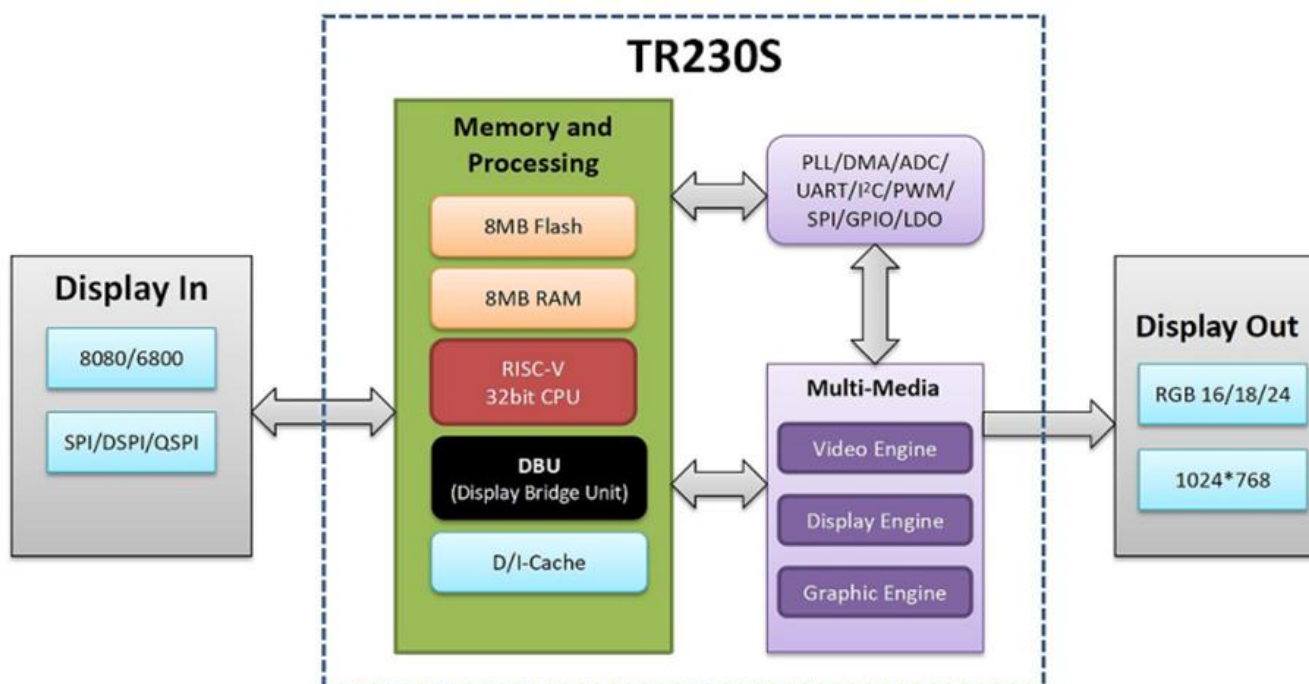
THIRD PARTY LICENCES MAY BE REQUIRED TO IMPLEMENT THE SOLUTION/PRODUCT. CUSTOMERS SHALL BE SOLELY RESPONSIBLE TO OBTAIN ALL APPROPRIATELY REQUIRED THIRD PARTY LICENCES. TDO SHALL NOT BE LIABLE FOR ANY LICENCE FEEOR ROYALTY DUE IN RESPECT OF ANY REQUIRED THIRD PARTY LICENCE. TDO SHALL HAVE NO WARRANTY, INDEMNITY OR OTHER OBLIGATIONS WITH RESPECT TO MATTERS COVERED UNDER ANY REQUIRED THIRD PARTY LICENCE.

## 1.1. GENERAL SPECIFICATION

- CPU
  - Single core E907, RV32IMAFC Instruction architecture, 400MHz@1.1V
  - D-Cache 32KB, I-Cache 16KB
  - PMP
  - CLINT and CLIC
- Embedded memory
  - SRAM 32KB
  - PSRAM 8MB
  - Flash 8MB
- Engine
  - DE (Display Engine):
    - One UI layer with 1024x768@60fps
    - Dithering, Gamma correction and color matrix adjustments
  - GE (Graphic Engine):
    - 2D graphics acceleration with max.1080P@60fps
    - Hardware rotation of 0, 90, 180, 270 degree
    - Hardware display mirroring
    - Hardware windowing
    - 1/16x ~ 16x scaling
  - VE (Video Engine):
    - JPEG decoder with max.720P@60fps
    - PNG decoder with max.720P@60fps
- Display Interface
  - Parallel RGB 16/18/24bits, Max. resolution 1024x768@60fps
- DBU(Display Bridge Unit) input
  - Support 8080 8/16bit parallel protocol
  - Support 6800 8/16bit parallel protocol
  - Support 3/4-wire SPI, DSPI, QSPI protocol
  - 8080/6800 8/16bit with max.50MHz clock
  - 3/4-wire SPI, 4-SDA SPI with max.100MHz clock
- Generic interfaces
  - SPIx2, 3WIRE /4WIRE
  - UARTx4, Supports 2-wire/3-wire/4-wire interfaces, compatible with industry standard 16550,
  - IIC x2, Max. rate 400Kb/s
  - Five GPIO groups with a total of 60 IOs, each IO can be configured independently:
    - The input supports secondary de-bounce and interrupt

- 
- Counters
    - GTC
    - WDOG
    - PWM
      - Supports up to four PWM outputs or two complementary PWM outputs with 16bit
  - Analog
    - 6 channel with 12-bit GPADC, sampling rate 1MSPS
    - Integrated RTP resistive touch screen interface
  - Clock and power management
    - Chip clock source
      - Crystal-less solution, using built-in OSC24M,  $\pm 2\%$  accuracy
    - CMU with PLLx4:
    - SYSCFG with LDOx3(embedded):
      - LDO25 (2.5V 100mA)
      - LDO18 (1.8V 100mA)
      - LDO1x (1.0~1.375V 200mA),
    - **3.3V single power needed only**
    - Built-in THS temperature sensor, support high and low temperature interrupt alarm and over-temperature reset chip
  - Package
    - QFN68, 7x7x0.85mm, pitch 0.35mm
  - Junction temperature
    - -20 to +105°C

## 1.2. TR230S BLOCK DIAGRAM

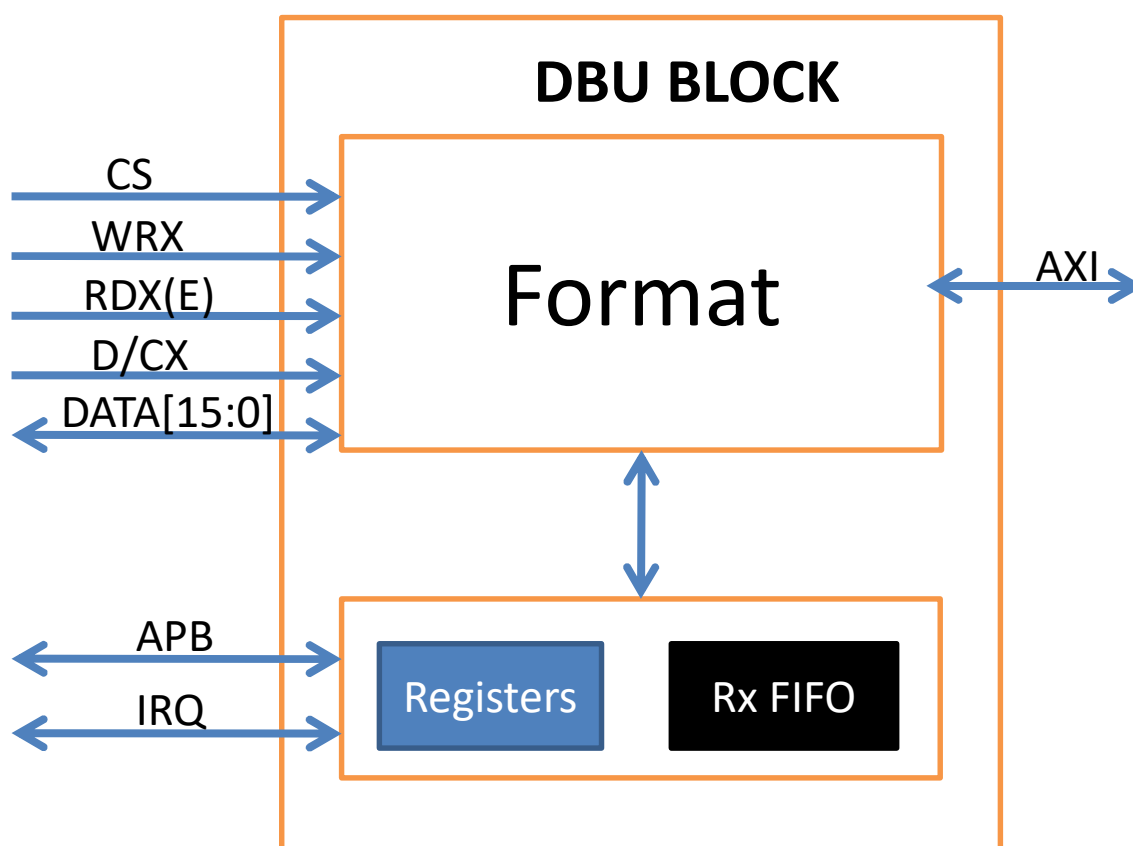




### 1.3. DBU BLOCK

The **DBU**(Display Bridge Unit)block is responsible for acquiring data from external MCUs and other systems, collecting image signals and outputting them to RAM, and then transferring them to DE for display. In this application scenario, the "external MCU" belongs to the host role, and the TR230S using the DBU is the device role, which is functionally equivalent to a "screen drive IC".

DBU Block Diagram:



## 1.4. PIN DEFINITION

### 1.4.1. DBU PIN

| Pin# | Symbol  | Type | PIN Description                                                            |                                                                                                           |
|------|---------|------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|      |         |      | 8080                                                                       | SPI/DSPI/QSPI                                                                                             |
| 48   | PB8     | IO   | WR#:WRITE strobe input                                                     | SPI-CLK/DSPI-CLK/QSPI-CLK                                                                                 |
| 49   | PB9     | IO   | RD#:READ strobe input                                                      | SPI-MOSI/DSPI-DATA0/QSPI-DATA0                                                                            |
| 50   | CS#     | I    | Chip selection, LOW active                                                 |                                                                                                           |
| 51   | D/C#    | I    | DATA or COMMAND flag,<br>D/C#="1": DATA or PARAMETER;<br>D/C#="0": COMMAND | DATA or COMMAND flag,<br>SPI-4WIRE interface needed:<br>D/C#="1": DATA or PARAMETER;<br>D/C#="0": COMMAND |
| 40   | PB0     | I    | 8080-Data0                                                                 |                                                                                                           |
| 41   | PB1     | IO   | 8080-Data1                                                                 | SPI-MISO/DSPI-DATA1/QSPI-DATA1                                                                            |
| 42   | PB2     | IO   | 8080-Data2                                                                 | QSPI-DATA2                                                                                                |
| 43   | PB3     | IO   | 8080-Data3                                                                 | QSPI-DATA3                                                                                                |
| 44   | PB4     | I    | 8080-Data4                                                                 |                                                                                                           |
| 45   | PB5     | I    | 8080-Data5                                                                 |                                                                                                           |
| 46   | PB6     | I    | 8080-Data6                                                                 |                                                                                                           |
| 47   | PB7     | I    | 8080-Data7                                                                 |                                                                                                           |
| 66   | DBU_D8  | I    | 8080-Data8                                                                 |                                                                                                           |
| 67   | DBU_D9  | I    | 8080-Data9                                                                 |                                                                                                           |
| 68   | DBU_D10 | I    | 8080-Data10                                                                |                                                                                                           |
| 1    | DBU_D11 | I    | 8080-Data11                                                                |                                                                                                           |
| 2    | DBU_D12 | I    | 8080-Data12                                                                |                                                                                                           |
| 3    | DBU_D13 | I    | 8080-Data13                                                                |                                                                                                           |
| 4    | DBU_D14 | I    | 8080-Data14                                                                |                                                                                                           |
| 5    | DBU_D15 | I    | 8080-Data15                                                                |                                                                                                           |
| 6    | PD27    | O    | LCD_DE                                                                     | LCD_DE                                                                                                    |
| 7    | PD26    | O    | LCD_VS                                                                     | LCD_VS                                                                                                    |
| 8    | PD25    | O    | LCD_HS                                                                     | LCD_HS                                                                                                    |
| 9    | PD24    | O    | LCD_DCLK                                                                   | LCD_DCLK                                                                                                  |
| 10   | PD23    | O    | LCD_D23                                                                    | LCD_D23                                                                                                   |
| 11   | PD22    | O    | LCD_D22                                                                    | LCD_D22                                                                                                   |
| 13   | PD21    | O    | LCD_D21                                                                    | LCD_D21                                                                                                   |
| 14   | PD20    | O    | LCD_D20                                                                    | LCD_D20                                                                                                   |
| 15   | PD19    | O    | LCD_D19                                                                    | LCD_D19                                                                                                   |
| 16   | PD18    | O    | LCD_D18                                                                    | LCD_D18                                                                                                   |
| 17   | PD17    | O    | LCD_D17                                                                    | LCD_D17                                                                                                   |
| 18   | PD16    | O    | LCD_D16                                                                    | LCD_D16                                                                                                   |
| 19   | PD15    | O    | LCD_D15                                                                    | LCD_D15                                                                                                   |
| 20   | PD14    | O    | LCD_D14                                                                    | LCD_D14                                                                                                   |
| 21   | PD13    | O    | LCD_D13                                                                    | LCD_D13                                                                                                   |
| 22   | PD12    | O    | LCD_D12                                                                    | LCD_D12                                                                                                   |
| 23   | PD11    | O    | LCD_D11                                                                    | LCD_D11                                                                                                   |
| 24   | PD10    | O    | LCD_D10                                                                    | LCD_D10                                                                                                   |
| 25   | PD9     | O    | LCD_D9                                                                     | LCD_D9                                                                                                    |
| 26   | PD8     | O    | LCD_D8                                                                     | LCD_D8                                                                                                    |

|    |     |   |        |        |
|----|-----|---|--------|--------|
| 30 | PD7 | 0 | LCD_D7 | LCD_D7 |
| 31 | PD6 | 0 | LCD_D6 | LCD_D6 |
| 32 | PD5 | 0 | LCD_D5 | LCD_D5 |
| 33 | PD4 | 0 | LCD_D4 | LCD_D4 |
| 34 | PD3 | 0 | LCD_D3 | LCD_D3 |
| 35 | PD2 | 0 | LCD_D2 | LCD_D2 |
| 36 | PD1 | 0 | LCD_D1 | LCD_D1 |
| 37 | PD0 | 0 | LCD_D0 | LCD_D0 |

## 1.4.2. POWER PIN

| Pin#       | Symbol    | Type | PIN Description                                                                             |
|------------|-----------|------|---------------------------------------------------------------------------------------------|
| 12, 29, 54 | VCC33_IO  | P    | External 3.3V POWER supply input for IO                                                     |
| 27, 53     | VDD11_SYS | 0    | Internal 1.xV LDO output, connect a 1uF decoupling cap.to GND for stabilization             |
| 28         | LD018     | 0    | Internal 1.8V LDO output, connect a 1uF and a 10uF decoupling cap.to GND for stabilization. |
| 55         | LD025     | 0    | Internal 2.5V LDO output, connect a 1uF decoupling cap.to GND for stabilization             |
| 69         | GND       | P    | GND(It is also a thermal pad)                                                               |

## 1.4.3. OTHER GPIOs

| Pin# | Symbol  | Type | PIN Description                                                                                                                                                                                               |
|------|---------|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 38   | BL-PWM  | 0    | PWM output for controlling backlight driver                                                                                                                                                                   |
| 39   | CTP-RST | IO   | RESET for CTP, LOW active: Connecting to CTP-RST when driving CTP(Capacitive Touch Panel) by TR230S.                                                                                                          |
| 52   | RESET#  | I    | RESET for TR230S and LCD. LOW active                                                                                                                                                                          |
| 56   | DBG_TXD | IO   | DBG_TXD:3.3VTTL UART_TRANSMIT, Reserving a test pad for updating F/W to TR230S, <b>a 10Kohm pull-up resistor is needed</b>                                                                                    |
| 57   | DBG_RXD | IO   | DBG_RXD:3.3VTTL UART_RECEIVE, Reserving a test pad for updating F/W to TR230S, <b>a 10Kohm pull-up resistor is needed</b>                                                                                     |
| 58   | GPADC2  | IO   | <b>LCDInit-SPICLK:</b> Connecting to panel when driving interface with RGB+SPI panel.<br><b>CTP-SCL:</b> Connecting to CTP-SCL when driving CTP(Capacitive Touch Panel) by TR230S(2.2K~4.7K pull-up needed).  |
| 59   | GPADC3  | IO   | <b>LCDInit-SPIDATA:</b> Connecting to panel when driving interface with RGB+SPI panel.<br><b>CTP-SDA:</b> Connecting to CTP-SDA when driving CTP(Capacitive Touch Panel) by TR230S(2.2K~4.7K pull-up needed). |
| 60   | GPADC4  | IO   | <b>LCDInit-CS:</b> Chip select, connecting to panel when driving interface with RGB+SPI panel.                                                                                                                |
| 61   | BUSY    | 0    | BUSY STATUS flag,<br>0: Sending Command Not Allowed<br>1: Sending Command Allowed                                                                                                                             |
| 62   | RTP_YP  | IO   | <b>RTP_YP:</b> Connecting to Resistive touch Panel_YP if driving RTP by TR230S.<br><b>IM0:</b> INTERFACE CONFIGURATION PIN                                                                                    |
| 63   | RTP_XP  | IO   | <b>RTP_XP:</b> Connecting to Resistive touch Panel_XP if driving RTP by TR230S.<br><b>IM1:</b> INTERFACE CONFIGURATION PIN                                                                                    |

|    |        |    |                                                                                                                                                                                                                                                                           |
|----|--------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 64 | RTP_XN | IO | <b>RTP_XN:</b> Connecting to Resistive touch Panel_XN if driving RTP by TR230S.<br><b>Interruption FLAG:</b> Connecting to user MCU when driving CTP(Capacitive Touch Panel) by TR230S;<br>Not connecting when user driving CTP(Capacitive Touch Panel) via IIC directly. |
| 65 | RTP_YN | IO | <b>RTP_YN:</b> Connecting to Resistive touch Panel_YN if driving RTP by TR230S.<br><b>CTP-INT:</b> Connecting to CTP-INT when driving CTP(Capacitive Touch Panel) by TR230S.                                                                                              |

Notes: I=INPUT, O=OUTPUT, IO= INPUT AND OUTPUT, P=POWER

#### 1.4.4. LCD INTERFACE PIN MAPPING

| PIN# | PIN Name | Mapping#   | 0     | 1     | 2     | 3     | 4     |
|------|----------|------------|-------|-------|-------|-------|-------|
|      |          | Bits/Pixel | 24    | 18    | 18    | 16    | 16    |
|      |          | Clks/Pixel | 1     | 1     | 1     | 1     | 1     |
| 37   | PD0      | LCD_D0     | B0    | NC    | B2    | NC    | B3    |
| 36   | PD1      | LCD_D1     | B1    |       | B3    |       | B4    |
| 35   | PD2      | LCD_D2     | B2    |       | B4    |       | B5    |
| 34   | PD3      | LCD_D3     | B3    |       | B5    |       | B6    |
| 33   | PD4      | LCD_D4     | B4    |       | B6    |       | B7    |
| 32   | PD5      | LCD_D5     | B5    |       | B7    |       | G2    |
| 31   | PD6      | LCD_D6     | B6    | B2    | G2    |       | G3    |
| 30   | PD7      | LCD_D7     | B7    | B3    | G3    |       | G4    |
| 26   | PD8      | LCD_D8     | G0    | B4    | G4    | B3    | G5    |
| 25   | PD9      | LCD_D9     | G1    | B5    | G5    | B4    | G6    |
| 24   | PD10     | LCD_D10    | G2    | B6    | G6    | B5    | G7    |
| 23   | PD11     | LCD_D11    | G3    | B7    | G7    | B6    | R3    |
| 22   | PD12     | LCD_D12    | G4    | G2    | R2    | B7    | R4    |
| 21   | PD13     | LCD_D13    | G5    | G3    | R3    | G2    | R5    |
| 20   | PD14     | LCD_D14    | G6    | G4    | R4    | G3    | R6    |
| 19   | PD15     | LCD_D15    | G7    | G5    | R5    | G4    | R7    |
| 18   | PD16     | LCD_D16    | R0    | G6    | R6    | G5    | NC    |
| 17   | PD17     | LCD_D17    | R1    | G7    | R7    | G6    |       |
| 16   | PD18     | LCD_D18    | R2    | R2    | NC    | G7    |       |
| 15   | PD19     | LCD_D19    | R3    | R3    |       | R3    |       |
| 14   | PD20     | LCD_D20    | R4    | R4    |       | R4    |       |
| 13   | PD21     | LCD_D21    | R5    | R5    |       | R5    |       |
| 11   | PD22     | LCD_D22    | R6    | R6    |       | R6    |       |
| 10   | PD23     | LCD_D23    | R7    | R7    |       | R7    |       |
| 9    | PD24     | LCD_DCLK   | DCLK  | DCLK  | DCLK  | DCLK  | DCLK  |
| 8    | PD25     | LCD_HS     | HSYNC | HSYNC | HSYNC | HSYNC | HSYNC |
| 7    | PD26     | LCD_VS     | VSNC  | VSNC  | VSNC  | VSNC  | VSNC  |
| 6    | PD27     | LCD_DE     | DE    | DE    | DE    | DE    | DE    |

---

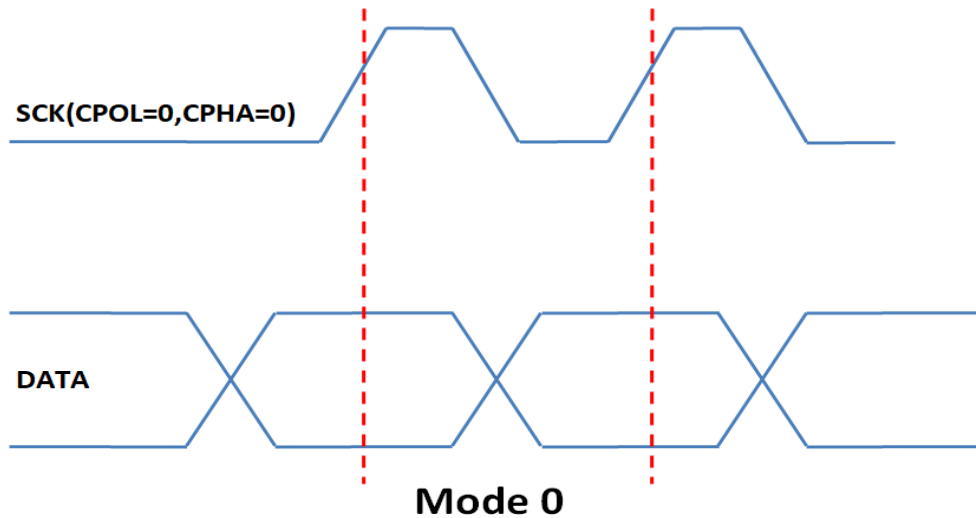
**Notes:**

There are 5 MAPPING output configurations, the default is 24bits/Pixel, which can be configured to 18bits and 16bits, and the corresponding low bits are not used.

- Mapping 0: Supports R/G/B signal interchange, and signal high-to-low sequence interchange within the group (Recommended using)
- Mapping 1: Supports the exchange of the entire set of R/G/B signals (Recommended using)
- Mapping 2: Supports R/G/B signal interchange
- Mapping 3: Supports R/G/B signal interchange (Recommended using)
- Mapping 4: Supports R/G/B signal interchange

## 1.5. INTERFACE DESCRIPTION

If there is no special indication, TR230S works on Mode0, that means data transmission occurs on the rising edge of the clock.



IM[1:0] Configuration table(6800 and DSPI is not commonly used and will not be described in this article):

| IM0 | IM1 | INTERFACE  | Related Pins                           | Max. Clock |
|-----|-----|------------|----------------------------------------|------------|
| 0   | 0   | SPI-4WIRE  | CS#, SCL, SD00, D/C#, BUSY             | 100MHz     |
| 1   | 0   | QSPI-4SDA  | CS#, SCL, SD00, SD01, SD02, SD03, BUSY | 100MHz     |
| 0   | 1   | 8080-8bit  | CS#, D0~D7, D/C#, RD#, WR#, BUSY       | 50MHz      |
| 1   | 1   | 8080-16bit | CS#, D0~D15, D/C#, RD#, WR#, BUSY      | 50MHz      |

### 1.5.1. SPI INTERFACE (SPI-4WIRE, QSPI-4SDA)

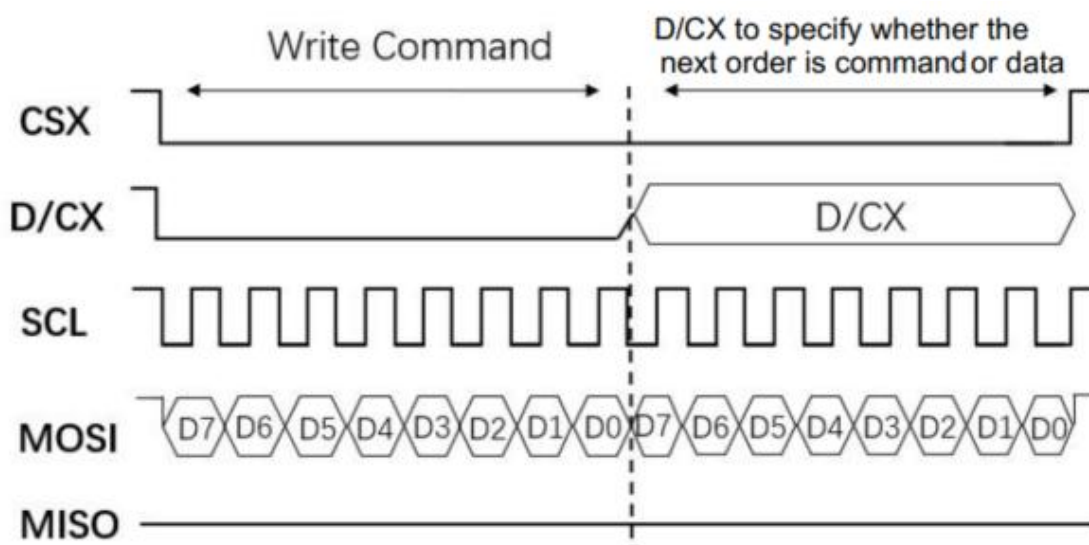
#### 1.5.1.1. PIN DESCRIPTION

| PIN NAME | Description                                                                     |
|----------|---------------------------------------------------------------------------------|
| CS#      | Chip selection: LOW active                                                      |
| SCL      | Serial Interface Clock                                                          |
| SD00     | SPI:MOSI<br>QSPI: QSPI-DATA0                                                    |
| SD01     | SPI:MISO<br>QSPI: QSPI-DATA1                                                    |
| SD02     | QSPI: QSPI-DATA2                                                                |
| SD03     | QSPI: QSPI-DATA3                                                                |
| D/C#     | DATA or COMMAND flag, 0:COMMAND 1:DATA                                          |
| BUSY     | BUSY STATUS flag,<br>0: Sending Command Not Allowed, 1: Sending Command Allowed |

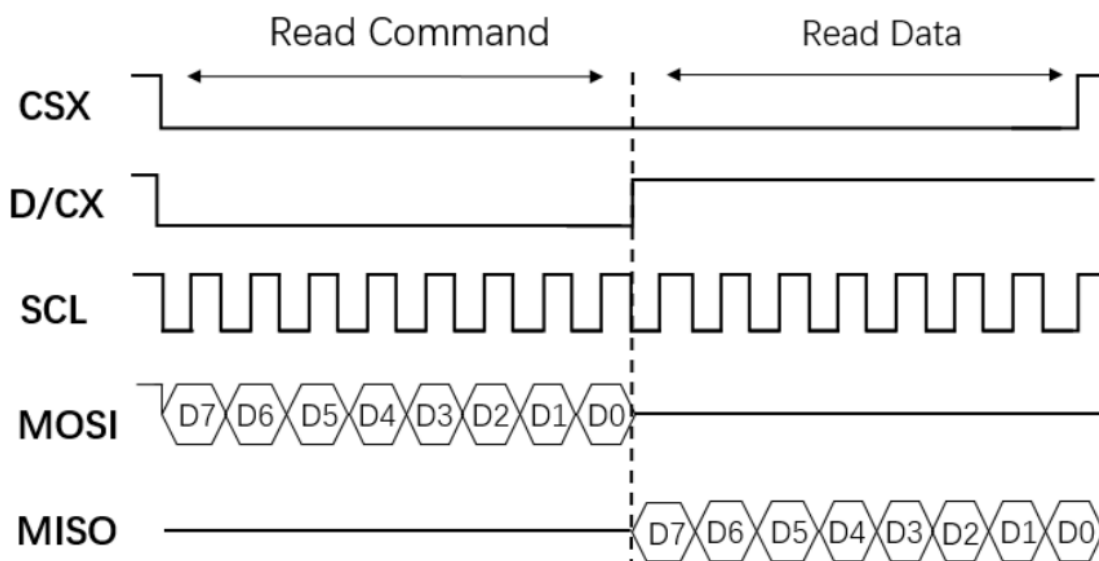
### 1.5.1.2. INTERFACE TIMING

- SPI-4WIRE

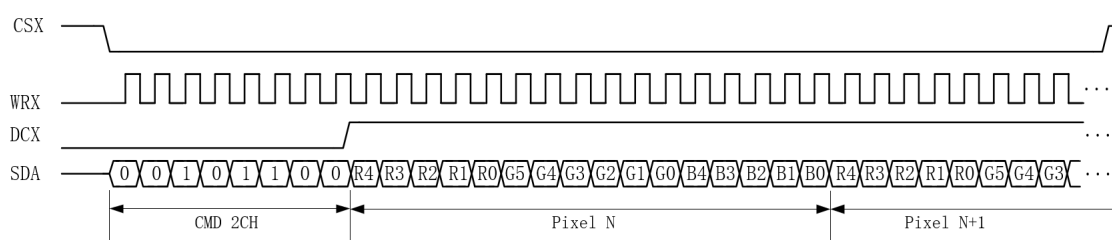
Timing for sending command and data:



Timing for reading:



Display data writing format:



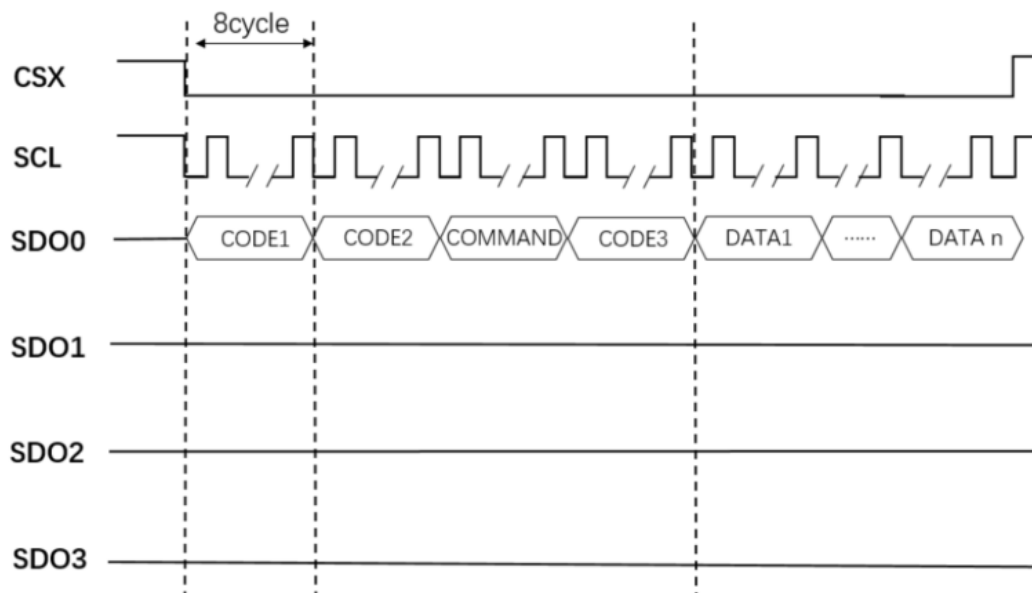
- QSPI-4SDA

Timing for writing COMMAND and PARAMETERS:

(When CODE1=0x02, CODE2=CODE3=0x00)

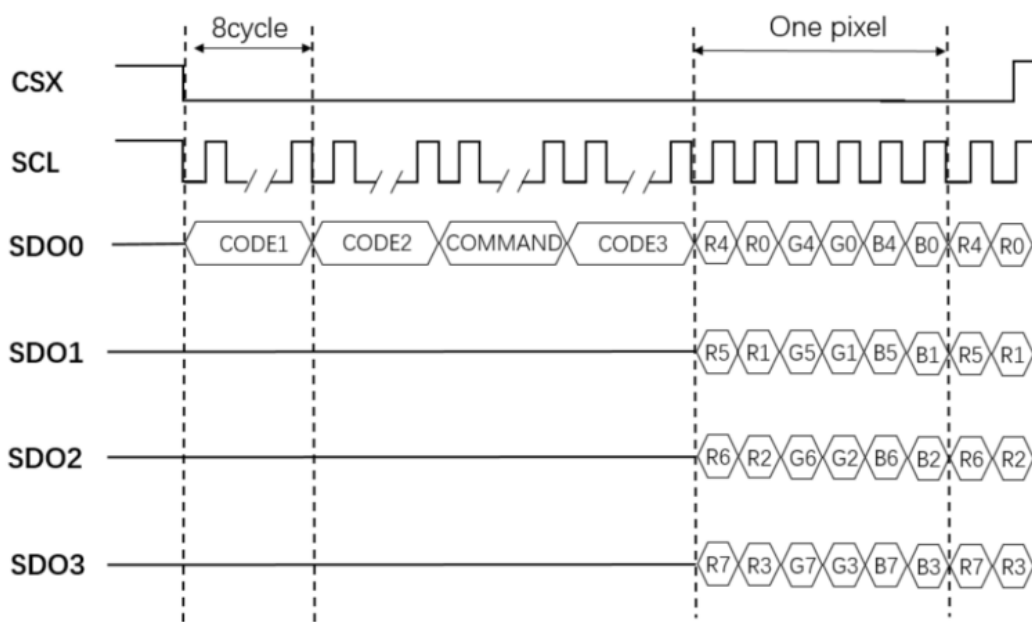
Timing for reading:

(When CODE1=0x03, CODE2=CODE3=0x00)



Display data writing format:

(CODE1=0x12, CODE2=CODE3=0x00 COMMAND = 0x2C)





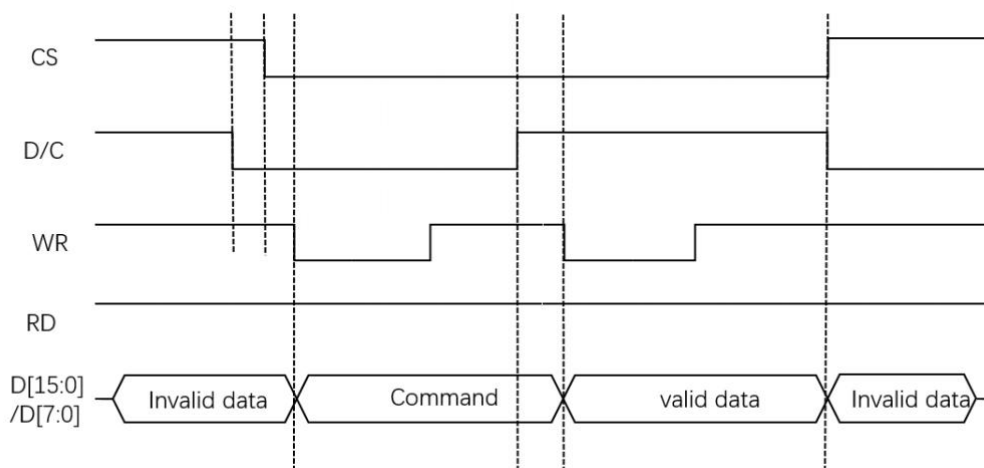
## 1.5.2. 8080 INTERFACE

### 1.5.2.1. PIN DESCRIPTION

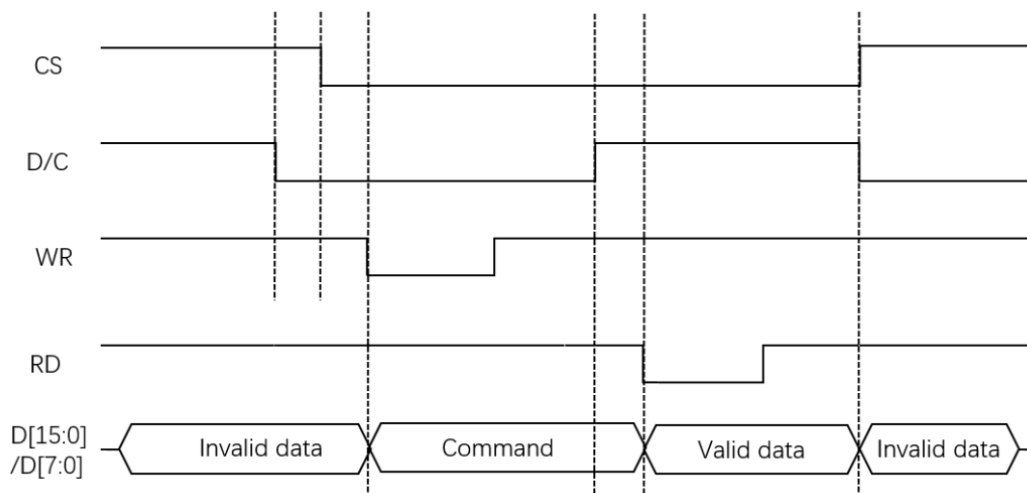
| PIN NAME | Description                                                                       |
|----------|-----------------------------------------------------------------------------------|
| CS#      | Chip selection: LOW active                                                        |
| WR       | WRITE strobe input, rising edge effective                                         |
| RD       | Read strobe input, rising edge effective                                          |
| DB0~DB15 | DATA BUS                                                                          |
| D/C#     | DATA or COMMAND flag, 0:COMMAND 1:DATA                                            |
| BUSY     | BUSY STATUS flag,<br>0: Sending Command Not Allowed<br>1: Sending Command Allowed |

### 1.5.2.2. INTERFACE TIMING (8080-8/16)

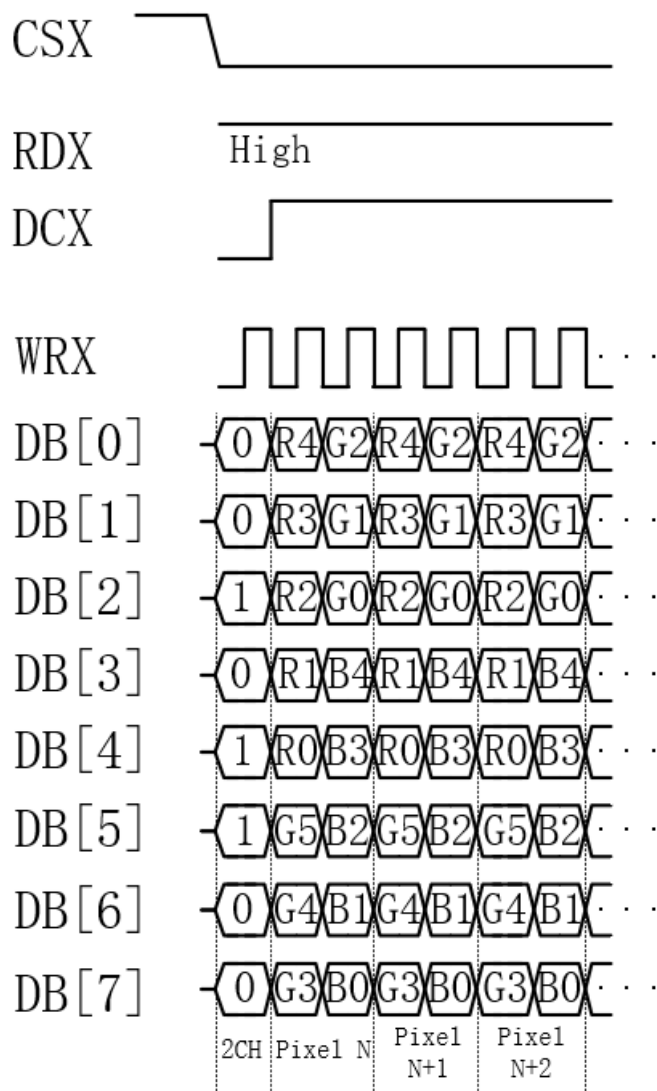
Timing for sending command and data:



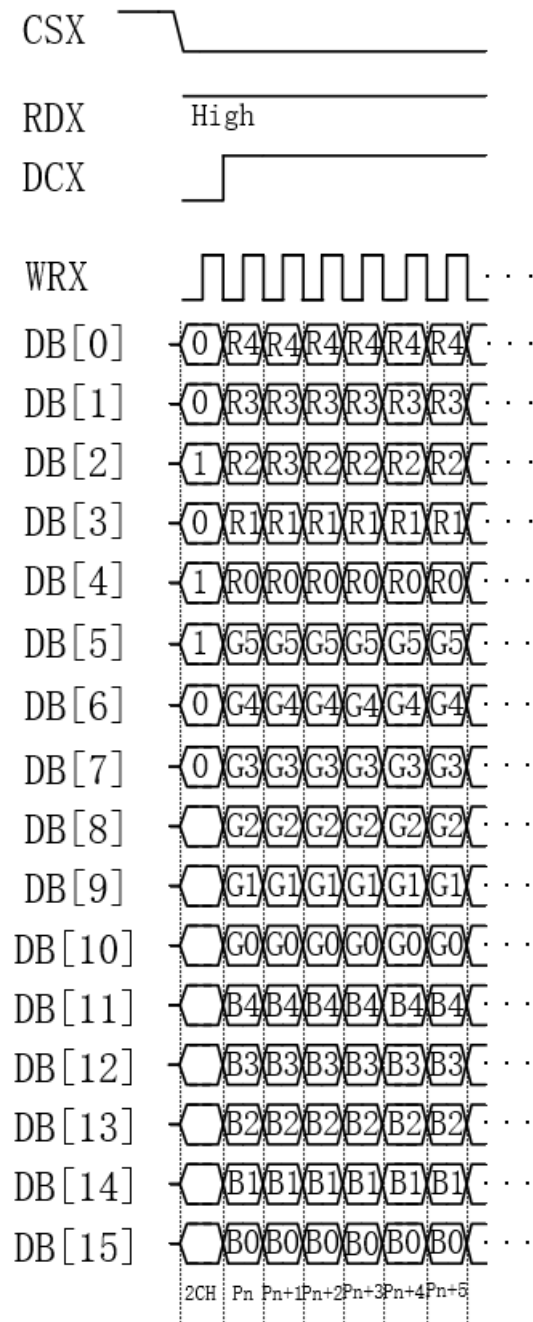
Timing for reading:



Display data writing format(8080-8B):



Display data writing format(8080-16B):



## 1.6. REGISTERS

### 1.6.1.Version (01H)

|             |              | Parameter |    |    |    |    |    |    |    |
|-------------|--------------|-----------|----|----|----|----|----|----|----|
|             | W/R          | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter | R            | H         |    |    |    | L  |    |    |    |
| Description | Version: H.L |           |    |    |    |    |    |    |    |

### 1.6.2.Self Test (12H)

|             |                                         | Parameter |    |    |    |    |    |    |    |     |
|-------------|-----------------------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                                     | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                                       | format    |    |    |    |    |    |    |    | 00H |
| Description | 0x00: Self test OFF; 0x01: Self test ON |           |    |    |    |    |    |    |    |     |

### 1.6.3.PWM Duty Ratio Setting (20H)

|             |                                                                            | Parameter |    |    |    |    |    |    |    |     |
|-------------|----------------------------------------------------------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                                                                        | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                                                                          | Duty      |    |    |    |    |    |    |    | 00H |
| Description | PWM duty ratio setting for backlight control: 0~100 (0x64 means 100% duty) |           |    |    |    |    |    |    |    |     |

### 1.6.4.PWM Frequency Setting (21H)

|             |                                                                            | Parameter |    |    |    |    |    |    |    |     |
|-------------|----------------------------------------------------------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                                                                        | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                                                                          | CLK       |    |    |    |    |    |    |    | 0AH |
| Description | PWM frequency setting for backlight control : 0~100KHz (0x64 means 100KHz) |           |    |    |    |    |    |    |    |     |

### 1.6.5.Display Off (28H)

|             |             | Parameter    |    |    |    |    |    |    |    |
|-------------|-------------|--------------|----|----|----|----|----|----|----|
|             | W/R         | D7           | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter | W           | No parameter |    |    |    |    |    |    |    |
| Description | Display off |              |    |    |    |    |    |    |    |

### 1.6.6.Display On (29H)

|             |            | Parameter    |    |    |    |    |    |    |    |
|-------------|------------|--------------|----|----|----|----|----|----|----|
|             | W/R        | D7           | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter | W          | No parameter |    |    |    |    |    |    |    |
| Description | Display on |              |    |    |    |    |    |    |    |

### 1.6.7.COL\_ADR (2AH)

|             |                            | Parameter |    |    |    |    |    |    |    |     |
|-------------|----------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                        | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                          | X_S (H)   |    |    |    |    |    |    |    | 00H |
| 2 Parameter |                            | X_S (L)   |    |    |    |    |    |    |    | 00H |
| 3 Parameter |                            | X_E (H)   |    |    |    |    |    |    |    | 00H |
| 4 Parameter |                            | X_E (L)   |    |    |    |    |    |    |    | 00H |
| Description | Indicates the display size |           |    |    |    |    |    |    |    |     |

### 1.6.8. ROW\_ADR (2BH)

|             |                            | Parameter |    |    |    |    |    |    |    |     |
|-------------|----------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                        | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                          | Y_S (H)   |    |    |    |    |    |    |    | 00H |
| 2 Parameter |                            | Y_S (L)   |    |    |    |    |    |    |    | 00H |
| 3 Parameter |                            | Y_E (H)   |    |    |    |    |    |    |    | 00H |
| 4 Parameter |                            | Y_E (L)   |    |    |    |    |    |    |    | 00H |
| Description | Indicates the display size |           |    |    |    |    |    |    |    |     |

### 1.6.9. DISPaly DATA (2CH)

|              |                                                                     | Parameter    |    |    |    |    |    |    |    |     |
|--------------|---------------------------------------------------------------------|--------------|----|----|----|----|----|----|----|-----|
|              | W/R                                                                 | D7           | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| Display data | W                                                                   | Display data |    |    |    |    |    |    |    |     |
| Description  | Display data push command (Special commands, Usage reference 1.5.1) |              |    |    |    |    |    |    |    |     |

### 1.6.10. DISPLAY FORMAT (3AH)

|             |                              | Parameter |    |    |    |    |    |    |    |     |
|-------------|------------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                          | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                            | format    |    |    |    |    |    |    |    | 55H |
| Description | RGB format for input signal: |           |    |    |    |    |    |    |    |     |
|             | 0x55:RGB565                  |           |    |    |    |    |    |    |    |     |
|             | 0x66:RGB666                  |           |    |    |    |    |    |    |    |     |
|             | 0x77:RGB888                  |           |    |    |    |    |    |    |    |     |

### 1.6.11. Soft Reset (5AH)

|             |            | Parameter |    |    |    |    |    |    |    |
|-------------|------------|-----------|----|----|----|----|----|----|----|
|             | W/R        | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter | W          | 0x01      |    |    |    |    |    |    |    |
| Description | Soft Reset |           |    |    |    |    |    |    |    |

### 1.6.12. RGB Interface Setting (70H)

|             |                                                                                                                                                                                                                                                                                                                                                                                                                 | Parameter   |    |    |    |    |    |    |    |     |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----|----|----|----|----|----|----|-----|
|             | W/R                                                                                                                                                                                                                                                                                                                                                                                                             | D7          | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                                                                                                                                                                                                                                                                                                                                                                                                               | Mode        |    |    |    |    |    |    |    | 01H |
| 2 Parameter |                                                                                                                                                                                                                                                                                                                                                                                                                 | Format      |    |    |    |    |    |    |    | 01H |
| 3 Parameter |                                                                                                                                                                                                                                                                                                                                                                                                                 | Clock_Phase |    |    |    |    |    |    |    | 01H |
| 4 Parameter |                                                                                                                                                                                                                                                                                                                                                                                                                 | Data_Order  |    |    |    |    |    |    |    | 00H |
| 5 Parameter |                                                                                                                                                                                                                                                                                                                                                                                                                 | Data_mirror |    |    |    |    |    |    |    | 00H |
| Description | 1>Mode<br>0x01:Parallel RGB<br>2>Format<br>0x01:RGB24bits(Mapping 0); 0x02:RGB18 with Low bits drop(Mapping 1); 0x03:RGB18 with high bits drop (Mapping 2); 0x04:RGB16 with Low bits drop (Mapping 3); 0x05:RGB16 with high bits drop (Mapping 4)<br>3>Clock_phase<br>0x01:0° ; 0x02:90° ; 0x03:180° ; 0x04:270°<br>4>Data_order<br>0x01:RGB; 0x02:RBG; 0x03:BGR; 0x04:BRG; 0x05:GRB; 0x06:GBR<br>5>Data_mirror |             |    |    |    |    |    |    |    |     |

|                                                                                 |
|---------------------------------------------------------------------------------|
| 0x00: Output RGB data from LSB to MSB;<br>0x01: Output RGB data from MSB to LSB |
|---------------------------------------------------------------------------------|

### 1.6.13. Resolution and Clock Setting (71H)

|             |                                                                                                         | Parameter  |    |    |    |    |    |    |    |     |
|-------------|---------------------------------------------------------------------------------------------------------|------------|----|----|----|----|----|----|----|-----|
|             | W/R                                                                                                     | D7         | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                                                                                                       | CLK        |    |    |    |    |    |    |    | 00H |
| 2 Parameter |                                                                                                         | Hactive(H) |    |    |    |    |    |    |    | 00H |
| 3 Parameter |                                                                                                         | Hactive(L) |    |    |    |    |    |    |    | 00H |
| 4 Parameter |                                                                                                         | Vactive(H) |    |    |    |    |    |    |    | 00H |
| 5 Parameter |                                                                                                         | Vactive(L) |    |    |    |    |    |    |    | 00H |
| Description | CLK: Pixel clock setting ,range with 0~48MHz; (0x30 means 48MHz)<br>Hactive/Vactive: Resolution setting |            |    |    |    |    |    |    |    |     |

### 1.6.14. Horizontal Porch Setting (72H)

|             |                          | Parameter              |    |    |    |    |    |    |    |     |
|-------------|--------------------------|------------------------|----|----|----|----|----|----|----|-----|
|             | W/R                      | D7                     | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                        | H-front porch (H)      |    |    |    |    |    |    |    | 00H |
| 2 Parameter |                          | H-front porch (L)      |    |    |    |    |    |    |    | 00H |
| 3 Parameter |                          | H-back porch (H)       |    |    |    |    |    |    |    | 00H |
| 4 Parameter |                          | H-back porch (L)       |    |    |    |    |    |    |    | 00H |
| 5 Parameter |                          | H-sync pulse width(H)  |    |    |    |    |    |    |    | 00H |
| 6 Parameter |                          | H-sync pulse width (L) |    |    |    |    |    |    |    | 00H |
| Description | Horizontal porch setting |                        |    |    |    |    |    |    |    |     |

### 1.6.15. Vertical Porch Setting (73H)

|             |                        | Parameter              |    |    |    |    |    |    |    |     |
|-------------|------------------------|------------------------|----|----|----|----|----|----|----|-----|
|             | W/R                    | D7                     | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                      | V-front porch (H)      |    |    |    |    |    |    |    | 00H |
| 2 Parameter |                        | V-front porch (L)      |    |    |    |    |    |    |    | 00H |
| 3 Parameter |                        | V-back porch (H)       |    |    |    |    |    |    |    | 00H |
| 4 Parameter |                        | V-back porch (L)       |    |    |    |    |    |    |    | 00H |
| 5 Parameter |                        | V-sync pulse width(H)  |    |    |    |    |    |    |    | 00H |
| 6 Parameter |                        | V-sync pulse width (L) |    |    |    |    |    |    |    | 00H |
| Description | Vertical porch setting |                        |    |    |    |    |    |    |    |     |

### 1.6.16. FONT Property Setting (75H)

|             |                                              | Parameter  |    |    |    |     |    |    |    |     |
|-------------|----------------------------------------------|------------|----|----|----|-----|----|----|----|-----|
|             | W/R                                          | D7         | D6 | D5 | D4 | D3  | D2 | D1 | D0 | Def |
| 1 Parameter | W                                            | Kerning(H) |    |    |    |     |    |    |    | 00H |
| 2 Parameter |                                              | Kerning(L) |    |    |    |     |    |    |    | 00H |
| 3 Parameter |                                              | Space (H)  |    |    |    |     |    |    |    | 00H |
| 4 Parameter |                                              | Space (L)  |    |    |    |     |    |    |    | 00H |
| 5 Parameter |                                              | F_E        |    |    |    | B_E |    |    |    | 00H |
| 6 Parameter |                                              | F_COLOR(H) |    |    |    |     |    |    |    | 00H |
| 7 Parameter |                                              | F_COLOR(L) |    |    |    |     |    |    |    | 00H |
| 8 Parameter |                                              | B_COLOR(H) |    |    |    |     |    |    |    | FFH |
| 9 Parameter |                                              | B_COLOR(L) |    |    |    |     |    |    |    | FFH |
| Description | Kerning: Font spacing<br>Space: Line spacing |            |    |    |    |     |    |    |    |     |

|  |                                                                                                                                                                                                |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | F_E: 0x0:Enable foreground color<br>0x1:Disable foreground color<br>B_E: 0x0:Enablebackground color<br>0x1: Disable background color<br>F_COLOR: Foreground color<br>B_COLOR: Background color |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 1.6.17. Host SPI function Enable (76H)

|             |                                                              | Parameter |    |    |    |    |    |    |    |     |
|-------------|--------------------------------------------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                                                          | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                                                            | format    |    |    |    |    |    |    |    | 00H |
| Description | Using SPI interface to send the initialization code of panel |           |    |    |    |    |    |    |    |     |
|             | 0x01: Enable Host SPI function                               |           |    |    |    |    |    |    |    |     |
|             | 0x00: Disable Host SPI function                              |           |    |    |    |    |    |    |    |     |

### 1.6.18. Host SPI Sending format (80H)

|               |                                                                                                                                                                                               | Parameter  |    |    |    |    |    |    |    |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----|----|----|----|----|----|----|
|               | W/R                                                                                                                                                                                           | D7         | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter   | W                                                                                                                                                                                             | CMD        |    |    |    |    |    |    |    |
| 2~N Parameter |                                                                                                                                                                                               | PARAMETERs |    |    |    |    |    |    |    |
| Description   | CMD: the first parameter is COMMOND register<br>2~N: the following are parameters of the COMMOND register<br>Indicates the format requirements for sending initialization code when using SPI |            |    |    |    |    |    |    |    |

### 1.6.19. 16\*16 Standard FONT library (81H)

|               |                                                                                                                 | Parameter |    |    |    |    |    |    |    |
|---------------|-----------------------------------------------------------------------------------------------------------------|-----------|----|----|----|----|----|----|----|
|               | W/R                                                                                                             | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter   | W                                                                                                               | X (H)     |    |    |    |    |    |    |    |
| 2 Parameter   |                                                                                                                 | X (L)     |    |    |    |    |    |    |    |
| 3 Parameter   |                                                                                                                 | Y (H)     |    |    |    |    |    |    |    |
| 4 Parameter   |                                                                                                                 | Y (L)     |    |    |    |    |    |    |    |
| 5~N Parameter |                                                                                                                 | String    |    |    |    |    |    |    |    |
| Description   | String: GBK character string with 16*16 dot matrix (ASCII characters displayed in a half width 8*16 dot matrix) |           |    |    |    |    |    |    |    |

### 1.6.20. 24\*24 Standard FONT library (82H)

|               |                                                                                                                   | Parameter |    |    |    |    |    |    |    |
|---------------|-------------------------------------------------------------------------------------------------------------------|-----------|----|----|----|----|----|----|----|
|               | W/R                                                                                                               | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter   | W                                                                                                                 | X (H)     |    |    |    |    |    |    |    |
| 2 Parameter   |                                                                                                                   | X (L)     |    |    |    |    |    |    |    |
| 3 Parameter   |                                                                                                                   | Y (H)     |    |    |    |    |    |    |    |
| 4 Parameter   |                                                                                                                   | Y (L)     |    |    |    |    |    |    |    |
| 5~N Parameter |                                                                                                                   | String    |    |    |    |    |    |    |    |
| Description   | String: GBK character string with 24*24 dot matrix（ASCII characters displayed in a half width 12 * 24 dot matrix） |           |    |    |    |    |    |    |    |

### 1.6.21. TP FUNCTION ENABLE (A0H)

|             |                              | Parameter |    |    |    |    |    |    |    |     |
|-------------|------------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                          | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                            | format    |    |    |    |    |    |    |    | 00H |
| Description | 0x00: Disable touch function |           |    |    |    |    |    |    |    |     |
|             | 0x01: Enable touch function  |           |    |    |    |    |    |    |    |     |

### 1.6.22. TP STATUS (A1H)

|             |                          | Parameter |    |    |    |    |    |    |    |     |
|-------------|--------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                      | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | R                        | TP_STATUS |    |    |    |    |    |    |    | 00H |
| Description | 0x00: No touching events |           |    |    |    |    |    |    |    |     |
|             | 0x01:Touching pressing   |           |    |    |    |    |    |    |    |     |
|             | 0x02:Touching Releasing  |           |    |    |    |    |    |    |    |     |

### 1.6.23. TP Coordinate (A2H)

|             |                      | Parameter |    |    |    |    |    |    |    |     |
|-------------|----------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                  | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | R                    | X (H)     |    |    |    |    |    |    |    | 00H |
| 2 Parameter |                      | X (L)     |    |    |    |    |    |    |    | 00H |
| 3 Parameter |                      | Y (H)     |    |    |    |    |    |    |    | 00H |
| 4 Parameter |                      | Y (L)     |    |    |    |    |    |    |    | 00H |
| Description | TP Coordinate (X, Y) |           |    |    |    |    |    |    |    |     |

### 1.6.24. Clearing TP STATUS (A3H)

|             |                          | Parameter |    |    |    |    |    |    |    |
|-------------|--------------------------|-----------|----|----|----|----|----|----|----|
|             | W/R                      | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter | W                        | Format    |    |    |    |    |    |    |    |
| Description | 0x00: To clear TP status |           |    |    |    |    |    |    |    |

### 1.6.25. Drawing LINE (B0H)

|              |                  | Parameter |    |    |    |    |    |    |    |
|--------------|------------------|-----------|----|----|----|----|----|----|----|
|              | W/R              | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter  | W                | X_S (H)   |    |    |    |    |    |    |    |
| 2 Parameter  |                  | X_S (L)   |    |    |    |    |    |    |    |
| 3 Parameter  |                  | Y_S (H)   |    |    |    |    |    |    |    |
| 4 Parameter  |                  | Y_S (L)   |    |    |    |    |    |    |    |
| 5 Parameter  |                  | X_E (H)   |    |    |    |    |    |    |    |
| 6 Parameter  |                  | X_E (L)   |    |    |    |    |    |    |    |
| 7 Parameter  |                  | Y_E (H)   |    |    |    |    |    |    |    |
| 8 Parameter  |                  | Y_E (L)   |    |    |    |    |    |    |    |
| 9 Parameter  |                  | Width     |    |    |    |    |    |    |    |
| 10 Parameter |                  | COLOR (H) |    |    |    |    |    |    |    |
| 11 Parameter |                  | COLOR (L) |    |    |    |    |    |    |    |
| Description  | Start (X S, Y S) |           |    |    |    |    |    |    |    |



|  |                                                         |
|--|---------------------------------------------------------|
|  | End(X_E, Y_E)<br>Width: Line width<br>COLOR: Line color |
|--|---------------------------------------------------------|

### 1.6.26. Drawing CIRCLE (B1H)

|             |                       | Parameter |    |    |    |    |    |    |    |
|-------------|-----------------------|-----------|----|----|----|----|----|----|----|
|             | W/R                   | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter | W                     | X (H)     |    |    |    |    |    |    |    |
| 2 Parameter |                       | X (L)     |    |    |    |    |    |    |    |
| 3 Parameter |                       | Y (H)     |    |    |    |    |    |    |    |
| 4 Parameter |                       | Y (L)     |    |    |    |    |    |    |    |
| 5 Parameter |                       | R (H)     |    |    |    |    |    |    |    |
| 6 Parameter |                       | R (L)     |    |    |    |    |    |    |    |
| 7 Parameter |                       | COLOR (H) |    |    |    |    |    |    |    |
| 8 Parameter |                       | COLOR (L) |    |    |    |    |    |    |    |
| Description | Center of circle(X,Y) |           |    |    |    |    |    |    |    |
|             | Radius (R)            |           |    |    |    |    |    |    |    |
|             | COLOR: Circle color   |           |    |    |    |    |    |    |    |

### 1.6.27. Drawing RECTANGLE (B2H)

|              |                                                                                                                                 | Parameter |    |    |    |    |    |    |    |
|--------------|---------------------------------------------------------------------------------------------------------------------------------|-----------|----|----|----|----|----|----|----|
|              | W/R                                                                                                                             | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 |
| 1 Parameter  | W                                                                                                                               | X1 (H)    |    |    |    |    |    |    |    |
| 2 Parameter  |                                                                                                                                 | X1 (L)    |    |    |    |    |    |    |    |
| 3 Parameter  |                                                                                                                                 | Y1 (H)    |    |    |    |    |    |    |    |
| 4 Parameter  |                                                                                                                                 | Y1 (L)    |    |    |    |    |    |    |    |
| 5 Parameter  |                                                                                                                                 | X2 (H)    |    |    |    |    |    |    |    |
| 6 Parameter  |                                                                                                                                 | X2 (L)    |    |    |    |    |    |    |    |
| 7 Parameter  |                                                                                                                                 | Y2 (H)    |    |    |    |    |    |    |    |
| 8 Parameter  |                                                                                                                                 | Y2 (L)    |    |    |    |    |    |    |    |
| 9 Parameter  |                                                                                                                                 | COLOR (H) |    |    |    |    |    |    |    |
| 10 Parameter |                                                                                                                                 | COLOR (L) |    |    |    |    |    |    |    |
| Description  | Coordinate of the upper left corner of the rectangle (X1,Y1),<br>Coordinate of the bottom right corner of the rectangle (X2,Y2) |           |    |    |    |    |    |    |    |

### 1.6.28. MIRROR AND ROTATION (ACH)

|             |                          | Parameter |    |    |    |    |    |    |    |     |
|-------------|--------------------------|-----------|----|----|----|----|----|----|----|-----|
|             | W/R                      | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | Def |
| 1 Parameter | W                        | format    |    |    |    |    |    |    |    | 00H |
| Description | 0x00:No Rotation         |           |    |    |    |    |    |    |    |     |
|             | 0x01:Rotation 90°        |           |    |    |    |    |    |    |    |     |
|             | 0x02: Rotation 180°      |           |    |    |    |    |    |    |    |     |
|             | 0x03: Rotation 270°      |           |    |    |    |    |    |    |    |     |
|             | 0x10:X-axis mirror image |           |    |    |    |    |    |    |    |     |
|             | 0x20:Y-axis mirror image |           |    |    |    |    |    |    |    |     |
|             | 0x40:Scaling             |           |    |    |    |    |    |    |    |     |

## 1.7. ELECTRICAL CHARACTERISTICS

### • Absolute Maximum Characteristics

| Symbol           | Description                         | Min  | Max | Unit |
|------------------|-------------------------------------|------|-----|------|
| T <sub>stg</sub> | Storage Temperature                 | -40  | 125 | °C   |
| I <sub>I/O</sub> | In/Out current for input and output | -50  | 60  | mA   |
| V <sub>I/O</sub> | I/O input voltage                   | -0.3 | 3.6 | V    |
| VCC              | Power Supply for VCC                | -0.3 | 3.6 | V    |

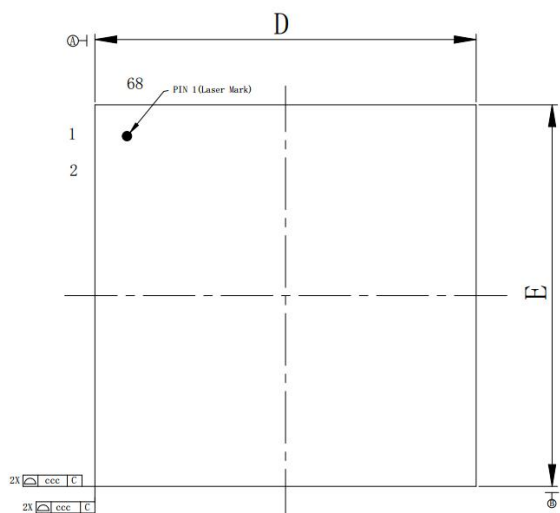
### • Recommended Operating Conditions

| Symbol         | Description                   | Min | Typ | Max | Unit |
|----------------|-------------------------------|-----|-----|-----|------|
| T <sub>a</sub> | Ambient Operating Temperature | -20 | -   | 85  | °C   |
| T <sub>j</sub> | Junction Temperature          | -20 |     | 105 | °C   |
| VCC            | Power Supply                  | 3.3 | 3.3 | 3.6 | V    |

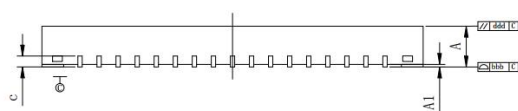
### • I/O Characteristics

| Symbol           | Description                      | Min       | Typ | Max       | Unit |
|------------------|----------------------------------|-----------|-----|-----------|------|
| V <sub>IH</sub>  | High-Level Input Voltage         | 0.7 * 3.3 | -   | 3.3 + 0.3 | V    |
| V <sub>IL</sub>  | Low-Level Input Voltage          | -0.3      | -   | 0.3 * 3.3 | V    |
| R <sub>PU</sub>  | Input pull-up resistance         |           | 33  |           | KΩ   |
| R <sub>PD</sub>  | Input pull-down resistance       |           | 33  |           | KΩ   |
| I <sub>IH</sub>  | High-Level Input Current         | -         | -   | 10        | uA   |
| I <sub>IL</sub>  | Low-Level Input Current          | -         | -   | 10        | uA   |
| V <sub>OH</sub>  | High-Level Output Voltage        | 3.3 - 0.2 | -   | 3.3       | V    |
| V <sub>OL</sub>  | Low-Level Output Voltage         | 0         | -   | 0.2       | V    |
| I <sub>OZ</sub>  | Tri-State Output Leakage Current | -10       | -   | 10        | uA   |
| C <sub>IN</sub>  | Input Capacitance                | -         | -   | 5         | pF   |
| C <sub>OUT</sub> | Output Capacitance               | -         | -   | 5         | pF   |

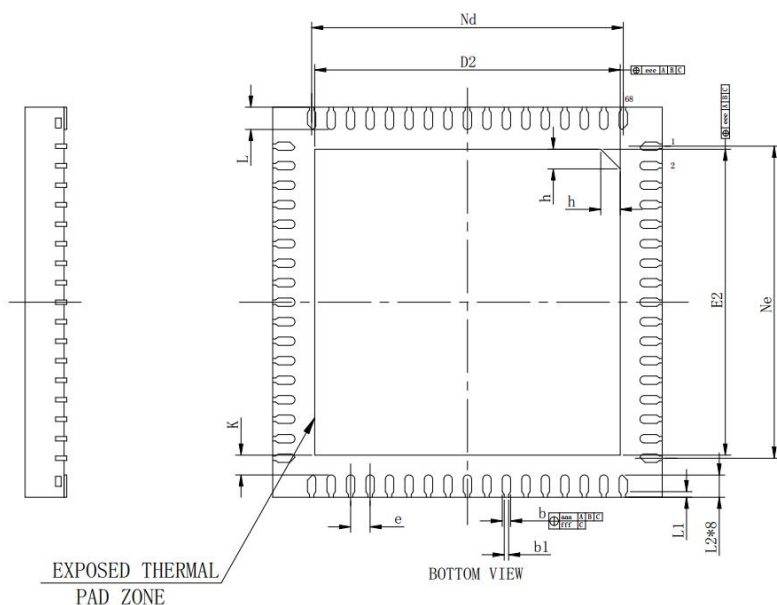
## 1.8. PACKAGE



TOP VIEW



SIDE VIEW



BOTTOM VIEW

| SYMBOL | MILLIMETER |      |      |
|--------|------------|------|------|
|        | MIN        | NOM  | MAX  |
| A      | 0.80       | 0.85 | 0.90 |
| A1     | —          | 0.02 | 0.05 |
| b      | 0.10       | 0.15 | 0.20 |
| b1     | 0.08REF    |      |      |
| c      | 0.18       | 0.20 | 0.25 |
| D      | 6.90       | 7.00 | 7.10 |
| D2     | 5.39       | 5.49 | 5.59 |
| e      | 0.35BSC    |      |      |
| Nd     | 5.60BSC    |      |      |
| E      | 6.90       | 7.00 | 7.10 |
| E2     | 5.39       | 5.49 | 5.59 |
| Ne     | 5.60BSC    |      |      |
| L      | 0.35       | 0.40 | 0.45 |
| L1     | 0.10REF    |      |      |
| L2     | 0.30       | 0.40 | 0.50 |
| K      | 0.20       | —    | —    |
| h      | 0.30       | 0.35 | 0.40 |
| aaa    | 0.07       |      |      |
| bbb    | 0.08       |      |      |
| ccc    | 0.10       |      |      |
| ddd    | 0.10       |      |      |
| eee    | 0.10       |      |      |
| fff    | 0.05       |      |      |

△